

AI Bathroom Designer

prompts i used to build it

Tools	OpenAI Codex (backend), Google Antigravity (frontend)
Repository	ai-bathroom-designer
Prepared	September 2026

1. Overview

You give it room size, budget and a style (or just a photo) and it gives back a priced design: 3D render, 2D CAD plan and a GST bill of materials in 3 tiers - Budget, Optimal and Luxury. Built mostly by prompting two coding assistants, one for each side.

Tool	Side	Built
Codex	Backend	Express + TypeScript API, constraint filter, scoring, GST bill of materials, optional photo step
Antigravity	Frontend	React + TypeScript app, 2D plan (Konva), 3D view (R3F), wizard, studio screen, chat copilot

2. How to run it

Open a terminal in the project folder and run:

```
cd ai-bathroom-designer
npm install
npm run dev:all
```

dev:all starts the API and the frontend together, so one terminal is enough.
Photo upload is optional. Put ANTHROPIC_API_KEY or GEMINI_API_KEY in a .env file to turn it on. No key = manual wizard, rest works the same.